PSYCH/COMPSCI/BMI 841: Computational Cognitive Science

Spring 2025
Prof. Joseph Austerweil

Guidelines for Project Proposals and Final Projects
Proposal Due March 12 at 8:59pm
Final Project Due May 6 at 8:59pm

Your projects are an opportunity to explore a question related to human and machine learning, following the spirit of the class and possibly applying some ideas from your readings. Different people have different backgrounds and interests, and you should draw on the strengths of yours when creating your project. You do not have to work on a project that relates directly to the topics covered in the classes and readings: other topics that pursue human and machine learning is fine, and you should try to work on a project that captures your interest. Working on some aspect of an existing research projects is okay (e.g., using some models from your class on your thesis data), as long as they are appropriate and your final paper is not identical to something you have turned in elsewhere. **You should meet with me before the project proposal is due to discuss your ideas.**

Note: If you are a graduate student, I will expect a bit more from your project. You should aim to produce something like a preliminary version of a conference paper (e.g., for the Cognitive Science Society Conference).

Categories of projects:

1. **Experiment.** Conduct a small experiment that tests a computational account of human cognition (probably one that was covered in the readings). As a guideline, a small experiment would be 10-20 people spending about 10 minutes each filling out a survey, or doing a simple computer task.
2. **Modeling or analysis.** Take some human behavior (experimental results from a paper we discussed in class or from one you read elsewhere, or “big data” from the Internet) and try to explain those results by implementing a formal model. You can create your own model or apply one from machine learning.
3. **Math.** Apply some of the mathematical ideas that we discussed to a particular example, or examine some consequences of a model we talked about in class.

Project proposal (due March 12 at 8:59pm)
Your proposal should be one page long (single-spaced), and should have four sections:

Background. Identify the topic your project will explore and briefly provide some of the context for your project. Describe what previous research has found in this area.
Question. State the specific question you are going to examine in your final project. This should only be one or two sentences.
Method. Briefly describe the method you are going to use to try to answer this question. Give some of the details behind your experimental procedure, your approach to modeling or analyzing the data, and/or your plans for analyzing the model.
References. Give at least three references cited in describing the background to your project.

You should submit your proposal through Canvas.

Project presentation (Class on 05/01)
Presentations will take place at the end of the semester. The goal of the presentations is to provide everyone the opportunity to see what the other people in the class have been up to in this semester. Your presentation should be at most ten minutes long. You should submit your presentation to Canvas immediately before or after class.

You should aim to communicate four things:

> **Background.** Provide the context for the class to understand your question.
> **Question.** The question you are going to examine in your final project.
> **Method.** The method you used (or are going to use) to answer this question.
> **Results.** Whatever (possibly preliminary) results you have obtained so far.

Your grade will be determined by the following factors:

> Appropriate discussion of Background, Question, Methods, and Results. Make sure to mention any implications and/or limitations of your Results (3.5 points)
> Presentation clarity (3 points)
> Appropriate use of time (1 point)

Please take a look at the Class Presentation Guide for useful tips for effective presentations.

Final project paper (due May 6 at 8:59pm)

Your paper should be about 6 pages single-spaced. There are no strict guidelines to the paper, but here is a suggestion. Start by motivating your research question, succinctly state your question, and outline the structure of the paper. Next, provide the background for your project, providing context to your question and discussing any previous work on it. Then, describe your methods and results, interpret and discuss the results, and briefly conclude. You are encouraged to cite papers in an APA-like manner, although any format is fine as long as it is clear and you are consistent.

The most important factor for your grade is to motivate and present the results of your project in a clear fashion. I will provide more precise criteria as we get closer to the due date.

You should submit your final paper through Canvas.